

Vendor Performance Analysis

Exploratory Data Analysis — uncovering patterns, anomalies, and opportunities across vendor transactions, margins, and inventory efficiency.

SUMMARY STATISTICS

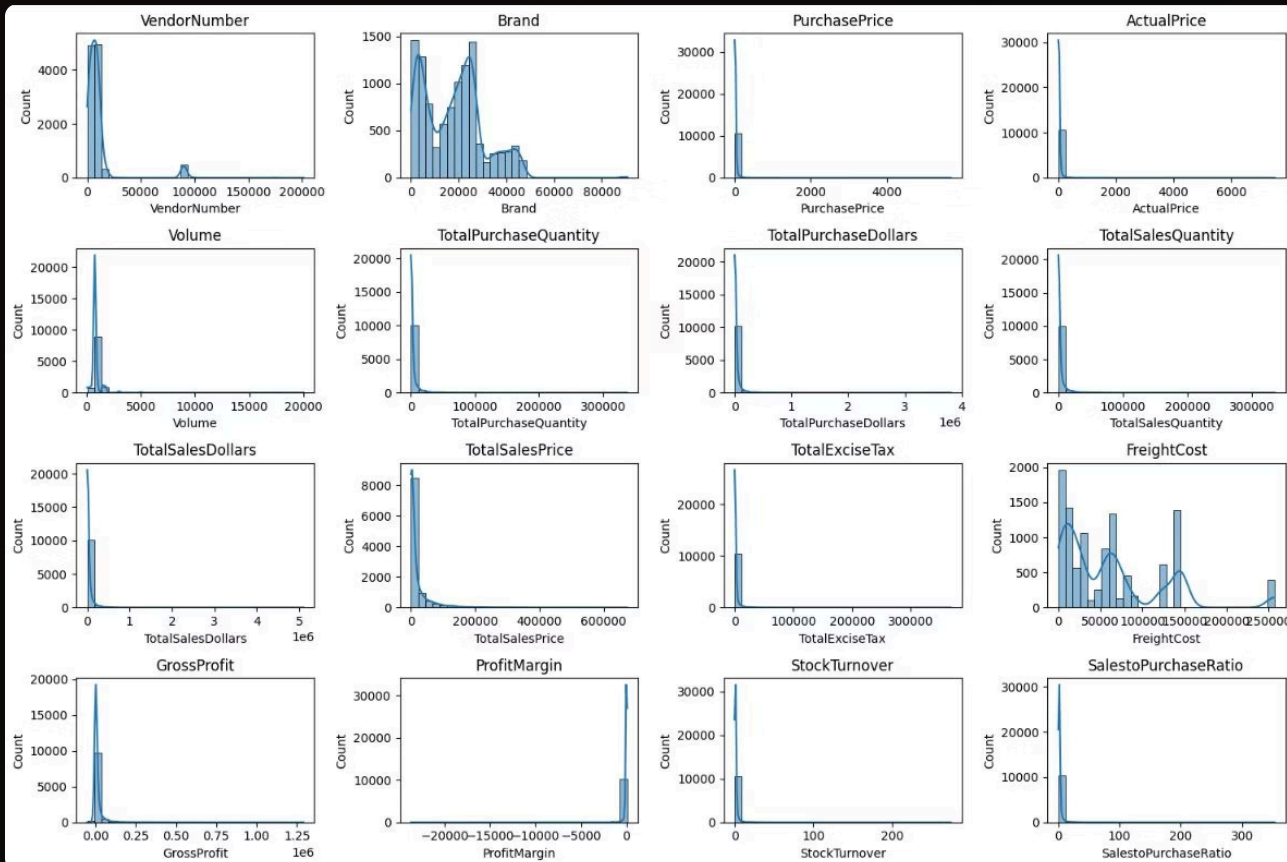
Key Metrics at a Glance

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalestoPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02

Notable Findings

- **Gross Profit** minimum of **-\$52,002** — some transactions result in outright losses
- **Profit Margin** extends to $-\infty$, signaling zero-revenue or deeply unprofitable sales
- **Freight Cost** ranges from \$0.09 to \$257,032 — extreme logistics cost variability
- **Stock Turnover** spans 0 to 274.5, revealing a wide gap between fast and stagnant SKUs

Histogram Analysis



What the Distributions Reveal

Most variables show **right-skewed distributions**, with a concentration of values near zero and long tails toward high extremes — typical of retail transaction data with a mix of high-volume staples and occasional premium items.

- ❏ Negative and zero-value bins confirm the presence of unprofitable and unsold inventory requiring data filtering before modeling.

DATA QUALITY

Negative & Zero Value Anomalies

Gross Profit

Minimum of **-\$52,002** — products sold below purchase cost due to heavy discounts or pricing errors.

Profit Margin

Extends to **$-\infty$** — revenue at or below total cost, producing undefined or extreme negative margins.

Sales Quantity & Dollars

Zero values indicate **slow-moving or obsolete stock** — purchased but never sold, creating inventory drag.

High-Variance Variables

Purchase & Actual Prices

Max values of **\$5,681** and **\$7,499** far exceed means of \$24 and \$35 — a small set of premium SKUs skews the distribution.

Freight Cost

Range of **\$0.09** to **\$257,032** suggests bulk shipments, logistics inefficiencies, or inconsistent carrier pricing across vendors.

Stock Turnover

Values from **0** to **274.5** — turnover above 1.0 means sales exceeded purchases, drawing on older inventory to fulfill demand.

Filtering for Reliable Insights

To ensure analysis reflects **genuinely profitable, active transactions**, three exclusion criteria were applied:

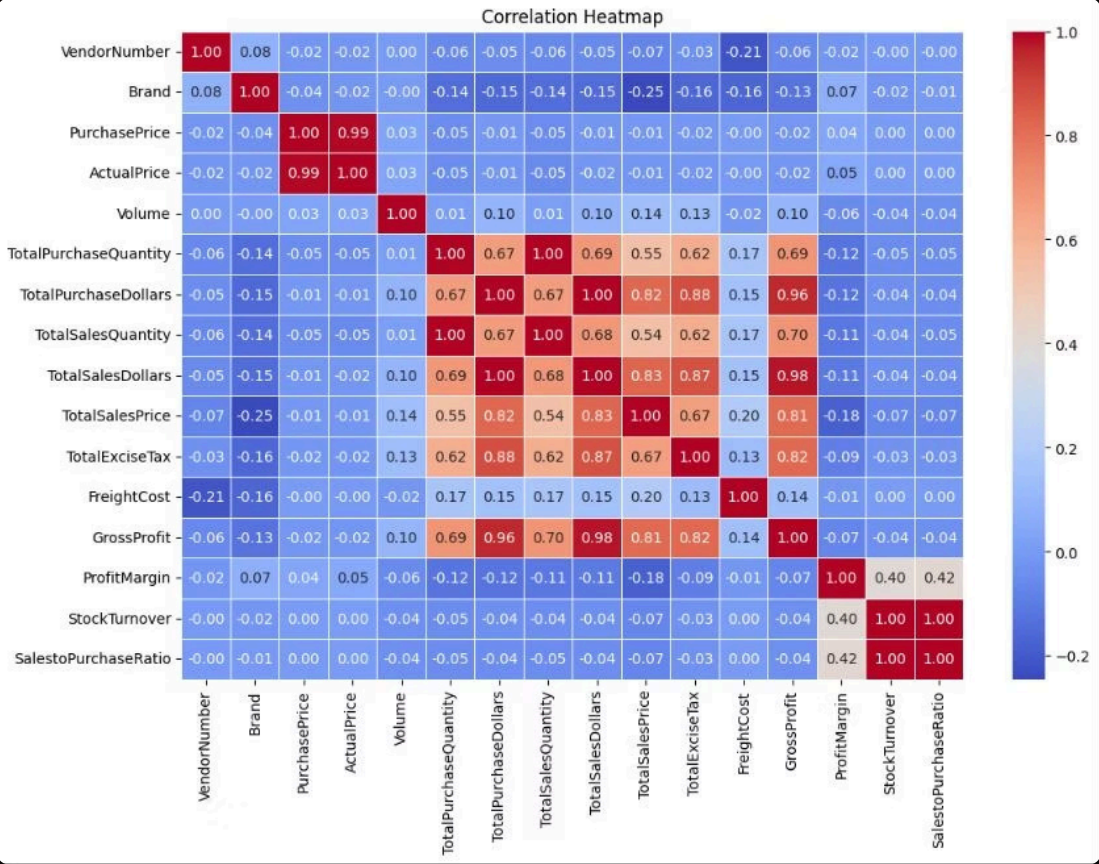
Remove Gross Profit ≤ 0

Remove Profit Margin ≤ 0

Remove Total Sales Quantity = 0

These filters remove noise from the dataset, focusing downstream correlation and vendor comparisons on transactions that represent real commercial activity.

Variable Relationships

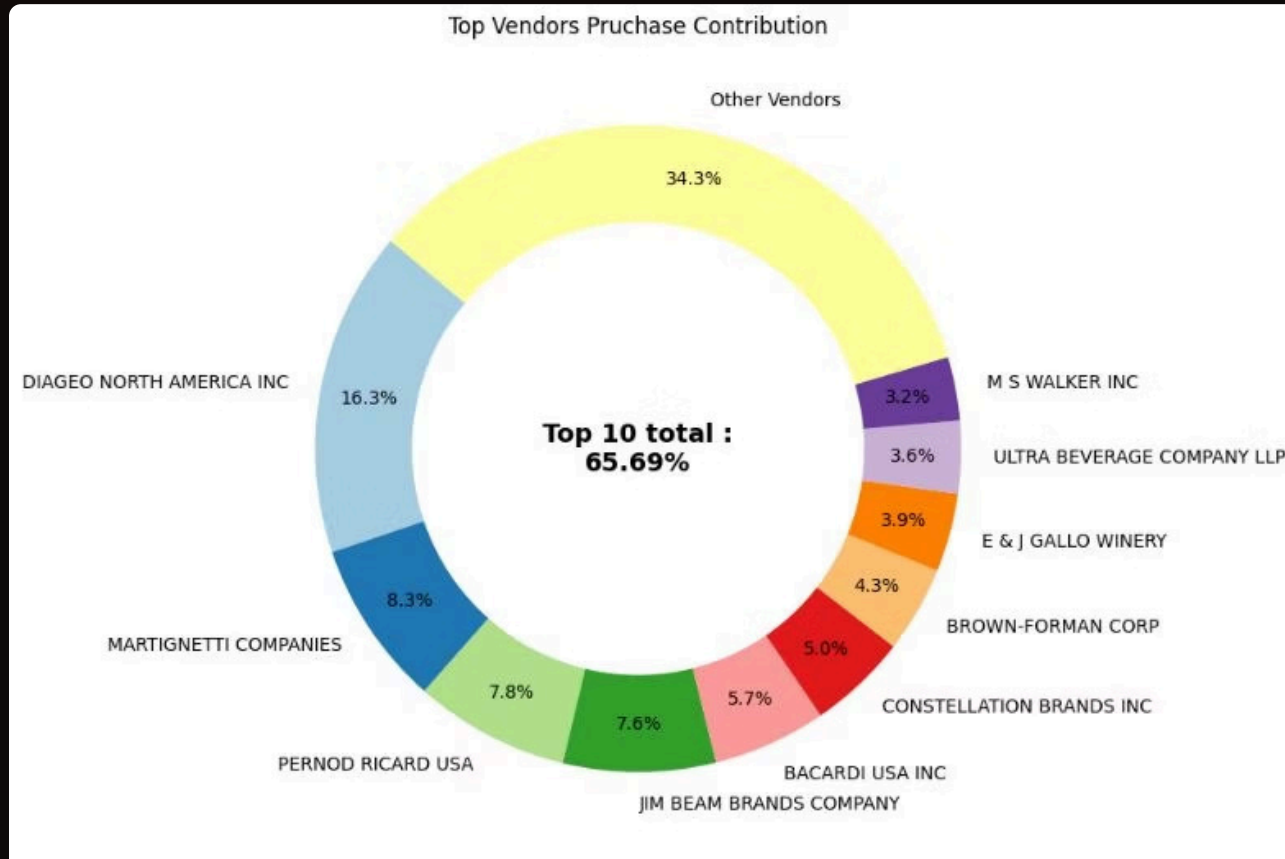


Key Correlations

- **Purchase Qty vs Sales Qty: 0.999** — near-perfect alignment confirms efficient inventory flow
- **Profit Margin vs Sales Price: -0.179** — higher prices may compress margins under competitive pressure
- **Purchase Price vs Gross Profit: -0.016** — price variation has negligible impact on profitability

i The near-zero correlations with purchase price suggest that margin drivers lie elsewhere — likely in freight, volume discounts, or vendor terms.

Top 10 Vendors Dominate Purchases



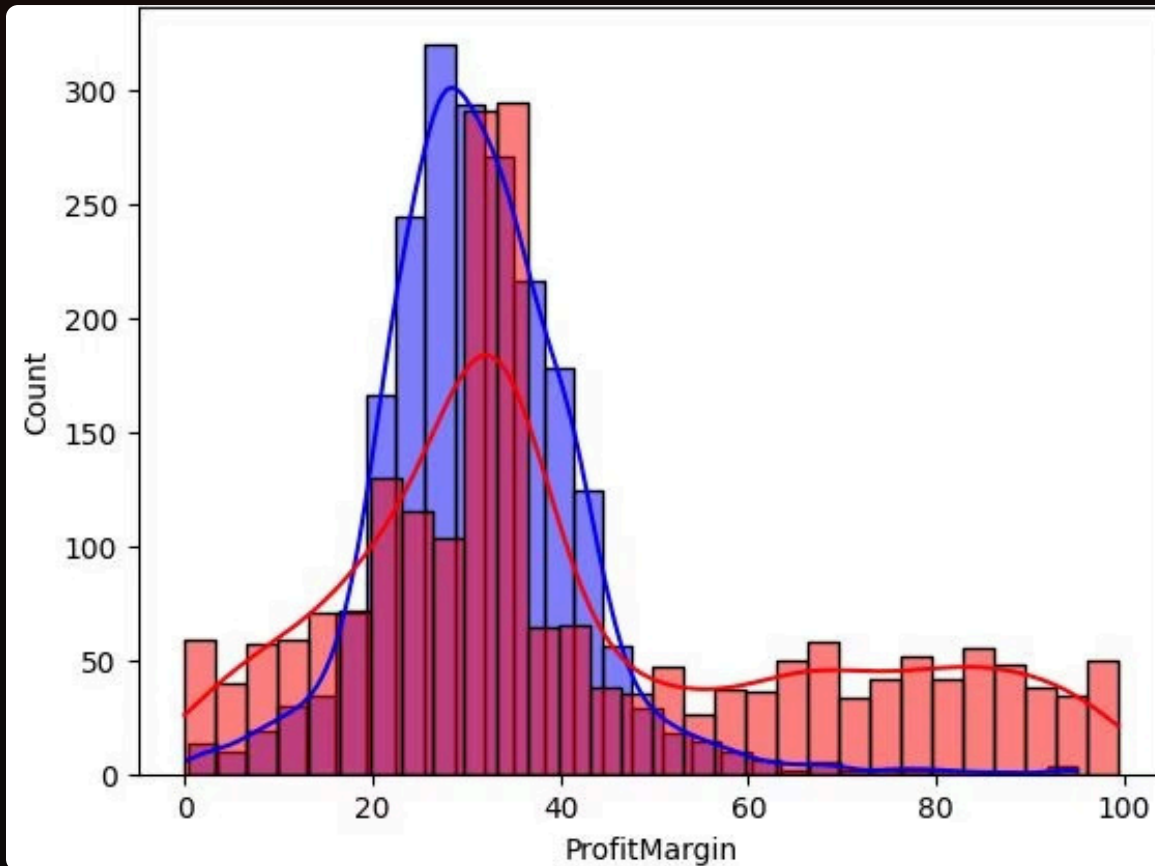
Concentration Risk

The top 10 vendors account for **65.69%** of total purchase volume, while all remaining vendors contribute just **32.31%**.

This heavy reliance creates **supply chain vulnerability** — disruptions from a single top vendor could materially impact operations.

⚠️ Diversifying the vendor base is a strategic priority to reduce concentration risk.

High vs. Low Performing Vendors



The Margin Paradox

Low-performing vendors maintain **higher margins** but struggle with volume — suggesting pricing inefficiencies or limited market reach rather than operational weakness.

Top Vendors

Mean margin: **31.17%** (95% CI: 30.74%–31.61%) — high volume, lower per-unit margin.

Low Vendors

Mean margin: **41.55%** (95% CI: 40.48%–42.62%) — higher margin, but weak sales volume.

Strategic Recommendations



Top-Performing Vendors

Optimize profitability by **matching pricing strategies**, reducing operational costs, and leveraging bulk ordering to improve margins without sacrificing volume.



Low-Performing Vendors

Improve **marketing efforts**, optimize pricing strategies, and enhance distribution networks to convert high margins into meaningful sales volume.



Vendor Diversification

Reduce concentration risk by **onboarding secondary vendors** to decrease reliance on the top 10 and strengthen supply chain resilience.